

PRINCIPLES OF BASEBAND DIGITAL DATA TRANSMISSION

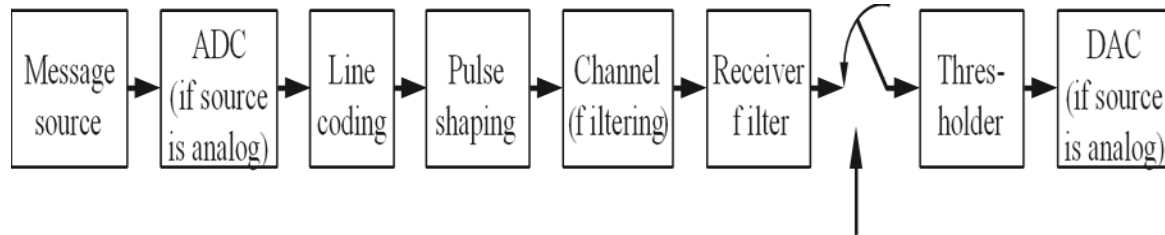


Figure 1

Synchronization

Figure1 shows a block diagram of a baseband digital data transmission system, which includes several possible signal processing operations.

As already mentioned, the analog-to-digital converter (ADC) block is present only if the source produces an analog message. It can be thought of as consisting of two operations--sampling and quantization.

The purposes of line coding are varied, and include spectral shaping, synchronization considerations, and bandwidth considerations, among other reasons.

Pulse shaping might be used to further shape the transmitted signal spectrum in order for it to be better accommodated by the transmission channel available.

At the output of the receiver filter, it is necessary to synchronize the sampling times to coincide with the received pulse epochs. The samples of the received pulses are then compared with a threshold in order to

make a decision as to whether a 0 or a 1 was sent (depending on the line code used, this may require some additional processing). If the data transmission system is operating reliably, these 1--0 decisions are correct with high probability and the resulting DAC output is a close replica of the input message waveform.

LINE CODES AND THEIR POWER SPECTRA

A digital signal is a sequence of discrete, discontinuous voltage pulses. Each pulse is a signal element. Binary data are transmitted by encoding each data bit into signal elements.

Definition of Terms

Unipolar: If the signal elements all have the same algebraic sign, that is, all positive or negative, then the signal is unipolar.

Polar signaling: In polar signaling, one logic state is represented by a positive voltage level, and the other by a negative voltage level.

The data signaling rate, or just data rate: Data rate of a signal is the rate, in bits per second, that data are transmitted. The duration or length of a bit is the amount of time it takes for the transmitter to emit the bit; for a data rate R , the bit duration is $1/R$.

Modulation Rate: The modulation rate, in contrast, is the rate at which the signal level is changed. This will depend on the nature of the digital encoding. The modulation rate is expressed in baud, which means signal elements per second.

Mark and Space: Finally, the terms mark and space, for historical reasons, refer to the binary digits 1 and 0, respectively.

The tasks involved in interpreting digital signals at the receiver can be summarized as follows:

1. The receiver must know the timing of each bit. That is, the receiver must know with some accuracy when a bit begins and ends.
2. The receiver must determine whether the signal level for each bit position is high (0) or low (1).

These tasks are performed by sampling each bit position in the middle of the interval and comparing the value to a threshold. Because of noise and other impairments.

What factors determine how successful the receiver will be in interpreting the incoming signal? Three factors are important:

- signal-to-noise ratio,
- the data rate
- the bandwidth.

With other factors held constant, the following statements are true:

- An increase in data rate increases bit error rate (BER).¹
- An increase in SNR decreases bit error rate.
- An increase in bandwidth allows an increase in data rate.

There is another factor that can be used to improve performance, and that is the encoding scheme. The encoding scheme is simply the mapping from data bits to signal elements.

Before describing these techniques, consider the following ways of evaluating or comparing the various encoding techniques:

- **Signal spectrum:** Several aspects of the signal spectrum are important. A lack of high-frequency components means that less bandwidth is required for transmission. In addition, lack of a direct-current (dc) component is also desirable. With a dc component to the signal, there must be direct physical attachment of transmission components. With no dc component, ac coupling via transformer is possible; this provides excellent electrical isolation, reducing interference. Finally, the magnitude of the effects of signal distortion and interference depend on the spectral properties of the transmitted signal. In practice, it usually happens that the transmission characteristics of a channel are worse near the band edges. Therefore, a good signal design should concentrate the transmitted power in the middle of the transmission bandwidth. In such a case, a smaller distortion should be present in the received signal. To meet this objective, codes can be designed with the aim of shaping the spectrum of the transmitted signal. The **Power spectrum** suitable for the particular channel available should be considered. For example, if the channel does not pass low frequencies, does the power spectrum of the chosen data format have a null at zero frequency? **Transmission bandwidth**---If the available transmission bandwidth is scarce, which it often is, a data format should be conservative in terms of bandwidth requirements. Sometimes

- **Clocking/Self synchronization:** We mentioned the need to determine the beginning and end of each bit position. This is no easy task. One rather expensive approach is to provide a separate clock lead to synchronize the transmitter and receiver. The alternative is to provide some synchronization mechanism that is based on the transmitted signal.

- **Error detection:** It is useful to have some error detection capability built into the physical signaling encoding scheme. This permits errors to be detected more quickly.

- **Signal interference and noise immunity:** Certain codes exhibit superior performance in the presence of noise. Performance is usually expressed in terms of a BER.

- **Cost and complexity:** Although digital logic continues to drop in price, this factor should not be ignored. In particular, the higher the signaling rate to achieve a given data rate, the greater the cost. We shall see that some codes require a signaling rate that is greater than the actual data rate. We now turn to a discussion of various techniques

Transparency : Every possible data sequence should be faithfully and transparently received, regardless of whether it is infrequent or not.

Line Code	Encoding rules
Nonreturn to Zero-Level (NRZ-L)	0=high level 1=low level
Nonreturn to Zero Inverted (NRZI)	0=no transition at beginning of interval 1=transition at beginning of interval
Bipolar-AMI	0=no line signal 1=positive or negative level, alternating for successive ones
Pseudoternary	0=positive or negative level, alternating for successive zeros 1=no line signal
Manchester	0=transition from high to low in middle of interval 1=transition from low to high in middle of interval

Differential Manchester	Always a transition in middle of interval 0=transition at beginning of interval 1=no transition at beginning of interval
HDB3	Same as bipolar AMI, except that any string of four zeros is replaced by a string with one code violation

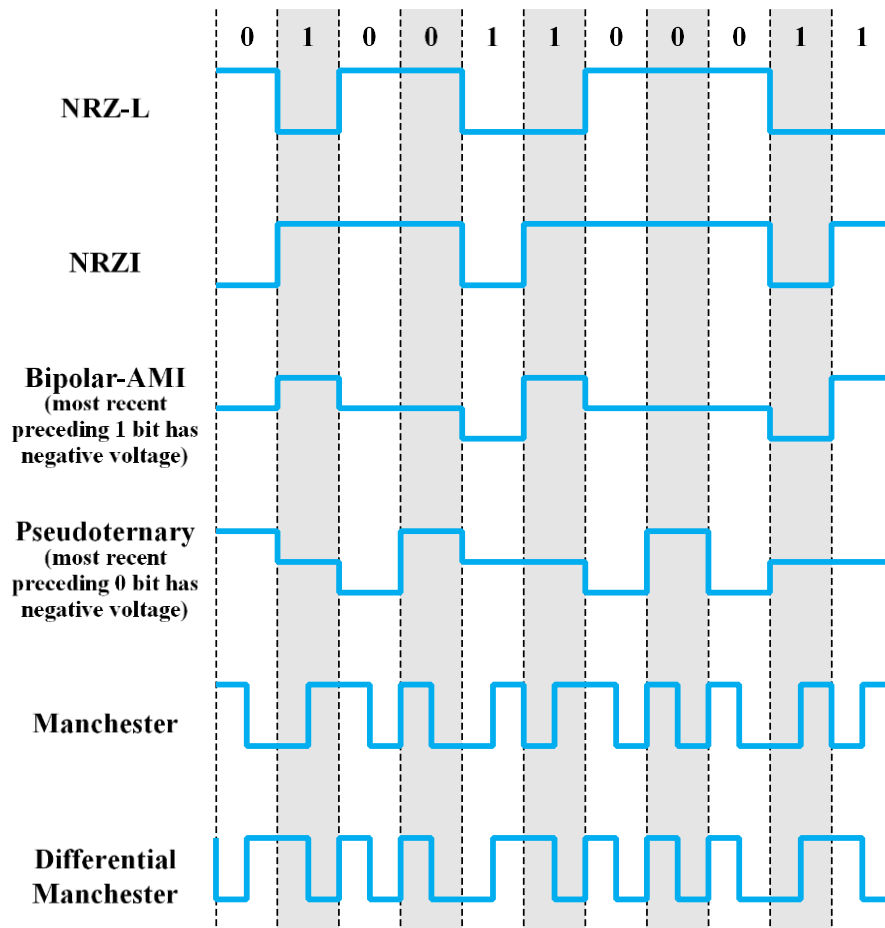


Figure 2

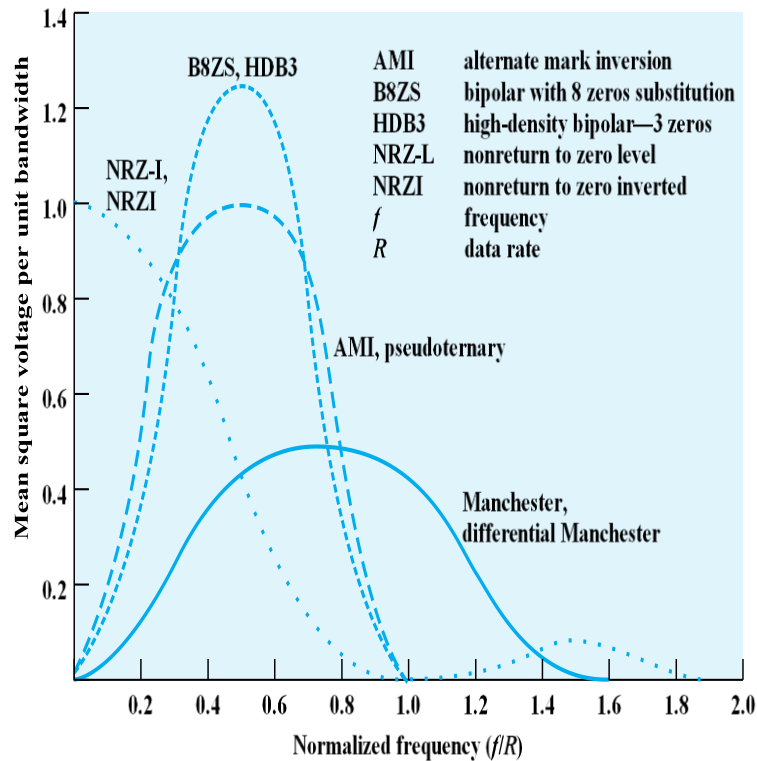


Figure 3

Multilevel Binary

A category of encoding techniques known as multilevel binary addresses some of the deficiencies of the NRZ codes. These codes use more than two signal levels. Two examples of this scheme are bipolar-AMI (alternate mark inversion) and pseudoternary

Biphase

There is another set of coding techniques, grouped under the term biphase, that overcomes the limitations of NRZ codes. Two of these techniques, Manchester and differential Manchester, are in common use. Differential Manchester has the added advantage of employing differential encoding.

All of the biphase techniques require at least one transition per bit time and may have as many as two transitions. Thus, the maximum

modulation rate is twice that for NRZ; this means that the bandwidth required is correspondingly greater.

On the other hand, the biphasic schemes have several advantages:

- **Synchronization:** Because there is a predictable transition during each bit time, the receiver can synchronize on that transition. For this reason, the biphasic codes are known as self-clocking codes.
- **No dc component:** Biphasic codes have no dc component, yielding the benefits described earlier
- **Error detection:** The absence of an expected transition can be used to detect errors. Noise on the line would have to invert both the signal before and after the expected transition to cause an undetected error.

As can be seen from Figure 3, the bandwidth for biphasic codes is reasonably narrow and contains no dc component. However, it is wider than the bandwidth for the multilevel binary codes. Biphasic codes are popular techniques for data transmission. The more common Manchester code has been specified for the IEEE 802.3 (Ethernet) standard for baseband coaxial cable and twisted-pair bus LANs. Differential Manchester has been specified for the IEEE 802.5 token ring LAN, using shielded twisted pair.

Modulation Rate

When signal-encoding techniques are used, a distinction needs to be made between data rate (expressed in bits per second) and modulation rate (expressed in baud). The data rate, or bit rate, is $\frac{1}{T_b}$ where T_b is the bit duration. The modulation rate is the rate at which signal elements are generated.

Consider, for example, Manchester encoding. The minimum size signal element is a pulse of one-half the duration of a bit interval. For a string of all binary zeroes or all binary ones, a continuous stream of such pulses is generated. Hence the maximum modulation rate for Manchester is $2/T_b$. In general $D = \frac{R}{L} = \frac{R}{\log_2 M}$

Where

$D = \text{Modulation rate, baud}$

$R = \text{Data rate, bps}$

$M = \text{number of signalling elements} = 2^L$

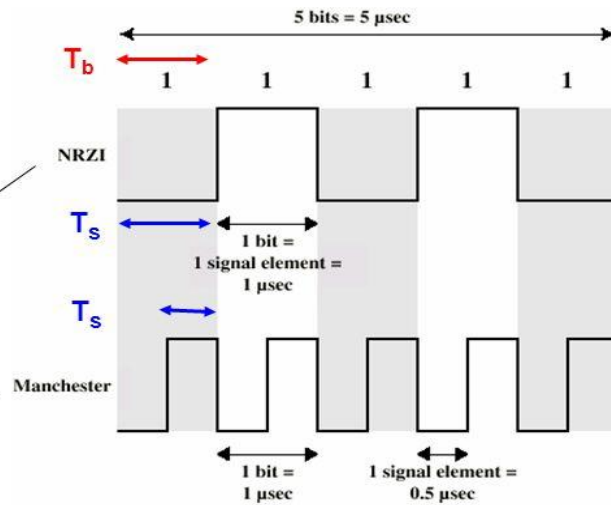
One way of characterizing the modulation rate is to determine the average number of transitions that occur per bit time. In general, this will depend on the exact sequence of bits being transmitted.

Example: Two different coding methods

Data rate = $1/1\mu\text{s}$
= 1 M **bps**

Signaling Rate for NRZI:
= $1/1\mu\text{s}$
= 1 M **bauds**

Signaling Rate for Manchester:
= $1/0.5\mu\text{s}$
= 2 M **bauds**



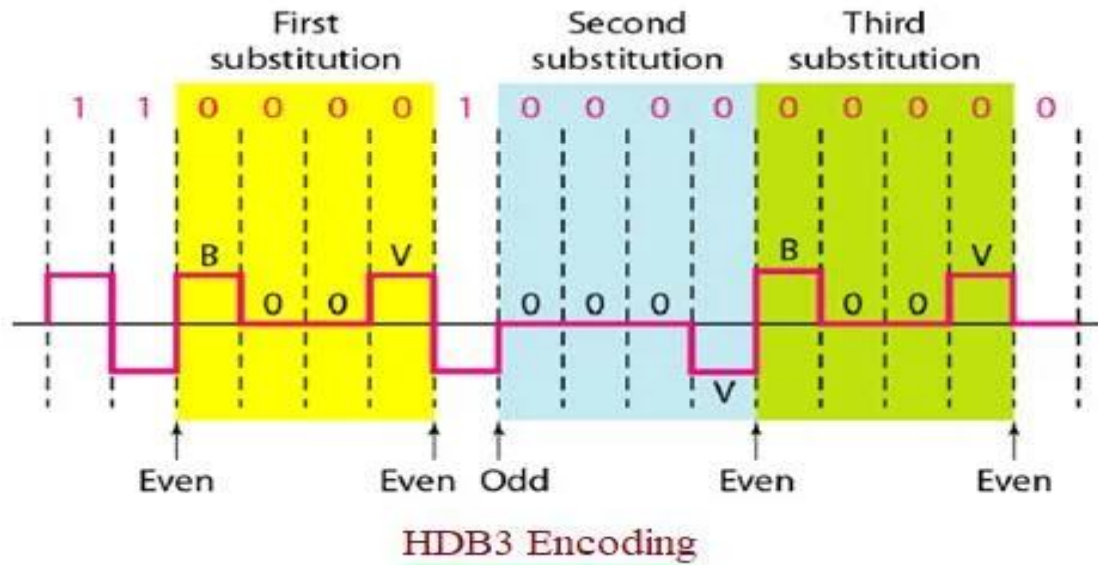
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HDB3 stands for High Density Bipolar 3 Zeros.

It is based on Bipolar AML. It is used in E1. It's a scrambling technique used in data communication to maintain data rate and to provide synchronization. In HDB3, four consecutive zero level voltages are replaced by sequence of "000V" or "B00V". These two substitutions are used to maintain even number of nonzero pulses.

The encoding is performed as per following two rules.

- If number of non-zero pulses after last substitution is odd, the substitution pattern is "000V", this makes total number of non-zero pulses even.
- If number of non-zero pulses after last substitution is even, the substitution pattern is "B00V", this maintains or keeps total number of non-zero pulses even.



Benefits Of HDB3 Encoding

Following are the benefits or **advantages of HDB3 encoding**:

- ➡ It substitutes long sequence of zero level pulses with combination of other levels to offer synchronization.
- ➡ It does not have any DC components in the encoded signal.
- ➡ It maintains total number of bits at input and output. Hence it does not increase or decrease the data rate.
- ➡ It offers built-in error detection capability.
- ➡ Most of the energy is concentrated around frequency which is equal to half of data rate.
- ➡ It is suitable for high data rate transmission over long distances.

Scrambling Techniques

Although the biphase techniques have achieved widespread use in local area network applications at relatively high data rates (up to 10 Mbps), they have not been widely used in long-distance applications. The principal reason for this is that they require a high signaling rate relative to the data rate. This sort of inefficiency is more costly in a long-distance application. Another approach is to make use of some sort of scrambling scheme. The idea behind this approach is simple: Sequences that would result in a constant voltage level on the line are replaced by filling sequences that will provide sufficient transitions for the receiver's clock to maintain synchronization. The filling sequence must be recognized by the receiver and replaced with the original data sequence. The filling sequence is the same length as the original sequence, so there is no data rate penalty. The design goals for this approach can be summarized as follows:

- No dc component
- No long sequences of zero-level line signals
- No reduction in data rate
- Error-detection capability